Guangdong Technion
Israel Institute of Technology
广东以色列理工学院

# GTIIT
# Calculus 1 - M
# Winter Semester - 2024

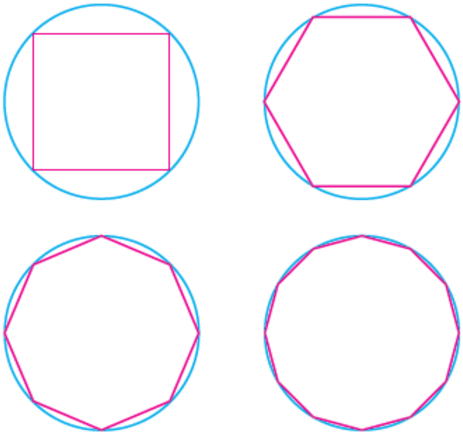
## Lecture 19 - December 16th
## 2nd Part

# ARC LENGTH

## Arc Length

- What do we mean by the length of a curve? We might think of fitting a piece of string to the curve and then measuring the string against a ruler. But that might be difficult to do with much accuracy if we have a complicated curve.
- We need a precise definition for the length of an arc of a curve, in the same spirit as the definitions we developed for the concepts of area and volume.
- If the curve is a polygon, we can easily find its length; we just add the lengths of the line segments that form the polygon: use the distance formula to find the distance between the endpoints of each segment.

# Arc Length



## Arc Length

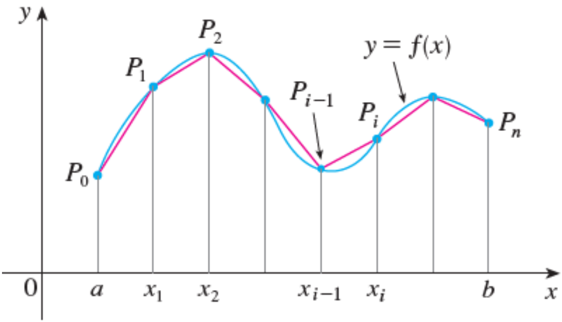
- We are going to find the length of a general curve by first approximating it by a polygon and then taking a limit as the number of segments of the polygon is increased.

- This process is familiar for the case of a circle, where the circumference is the limit of lengths of inscribed polygons.

## Arc Length

- Now suppose that a curve $C$ is defined by the equation $y = f(x)$, where $f$ is continuous and $a < x < b$.
- We obtain a polygonal approximation to $C$ by dividing the interval $[a, b]$ into $n$ subintervals with endpoints $x_0, x_1, \ldots, x_n$ and equal width $\Delta x$.
- If $y_k = f(x_k)$, then the point $P_k = (x_k, y_k)$ lies on $C$ and the polygon with vertices $P_0, P_1, \ldots, P_n$, is an approximation to $C$.
- The length $L$ of $C$ is approximately the length of this polygon and the approximation gets better as we let $n$ increase.

## Arc Length



## Arc Length

We define the length $L$ of the curve $C$ with equation $y = f(x)$, where $f$ is continuous and $a < x < b$, as the limit of the lengths of these inscribed polygons, provided the limit exists:

$$L = \lim_{n \to \infty} \sum_{i=1}^{n} \left\| P_k - P_{k-1} \right\| = \lim_{n \to \infty} \sum_{k=1}^{n} \sqrt{(x_k - x_{k-1})^2 + (y_k - y_{k-1})^2}$$

$$= \lim_{n \to \infty} \sum_{k=1}^{n} \sqrt{(\Delta x)^2 + (\Delta y_k)^2}$$

## Arc Length

For computational purposes, but we can derive an integral formula for $L$ in the case where f has a continuous derivative:
Such a function f is called smooth because a small change in $x$ produces a small change in $f'(x)$

Arc Length $L = \lim_{n\to\infty} \sum_{k=1}^{n} \sqrt{(\Delta x)^2 + (\Delta y_k)^2}$

By applying the Mean Value Theorem to $f$ on the interval $[x_{k-1}, x_k]$, we find that there is a number $x_k^*$ on $[x_{k-1}, x_k]$ such that

$$f(x_k) - f(x_{k-1}) = f'(x_k^*)(x_k - x_{k-1})$$

Then, we obtain

$$\Delta y_k = f'(x_k^*)(\Delta x)$$

$$\implies \sqrt{(\Delta x)^2 + (\Delta y_k)^2} = \sqrt{(\Delta x)^2 + [f'(x_k^*)(\Delta x)]^2}$$

$$= \sqrt{(\Delta x)^2}\sqrt{1 + [f'(x_k^*)]^2} = \Delta x\sqrt{1 + [f'(x_k^*)]^2}$$

## Arc Length

If $f'$ is continuous on $[a, b]$, then the length of the curve $y = f(x)$ on $[a, b]$ is

$$\boxed{L = \lim_{n\to\infty} \sum_{k=1}^{n} \Delta x \sqrt{1 + [f'(x_k^*)]^2}}$$

We recognize this as the limit of a Riemann Sum

$$\boxed{L = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx}$$

In Leibniz notation for differentials:

$$\boxed{L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx}$$

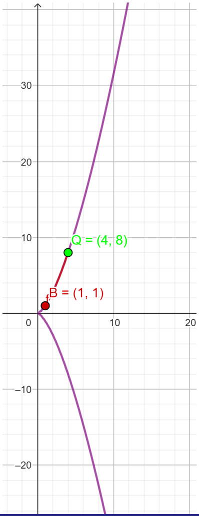
## Arc Length

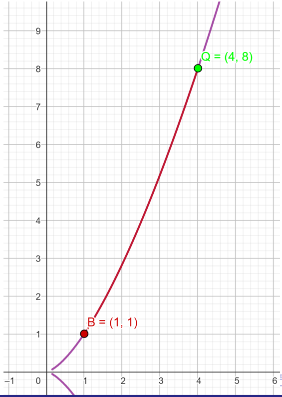
If a curve has the equation $x = g(y), c < y < d$, and $g'(y)$ is continuous, then by interchanging the roles of $x$ and $y$ in the previous formula, we obtain the following formula for its length:

$$\boxed{L = \int_c^d \sqrt{1 + [g'(y)]^2}\, dy}$$

$$\boxed{L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy}$$

## Arc Length Ex 1. Find the length of the semicubical parabola $y^2 = x^3$ between the points $(1,1)$ and $(4,8)$.



For the top half of the curve, we can solve $y = f(x)$ but better, implicit differentiation:

$$y^2 = x^3 \Longrightarrow 2y\frac{dy}{dx} = 3x^2 \Longrightarrow \frac{dy}{dx} = \frac{3}{2}\frac{x^2}{y} \Longrightarrow \left(\frac{dy}{dx}\right)^2 = \frac{9}{4}\frac{x^4}{y^2} = \frac{9}{4}x$$

And so, the arc length has the value

$$\boxed{L = \int_a^b \sqrt{1 + [f'(x)]^2 dx}}$$

$$\boxed{= \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_1^4 \sqrt{1 + \frac{9x}{4}}\, dx}$$

## Solve the integral by Substitution

- $u = 1 + \frac{9}{4}x \implies du = \frac{9}{4}\,dx$

- For the endpoints:

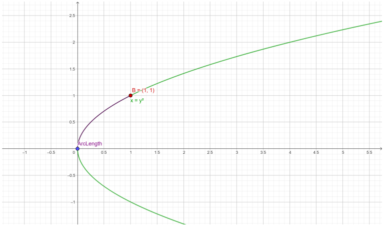
$x = 1 \implies u = \frac{13}{4}\ and\ x = 4 \implies u = 10$

Therefore, the arc length is

$$\boxed{L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx = \int_1^4 \sqrt{1 + \frac{9}{4}x}\,dx = \int_{\frac{13}{4}}^{10} \sqrt{u}\ \frac{4}{9}\,du}$$

$$= \frac{4}{9}\ \frac{2}{3}\ u^{\frac{3}{2}}\Big|_{\frac{13}{4}}^{10} = \frac{8}{27}\left[10^{\frac{3}{2}} - \left(\frac{13}{4}\right)^{\frac{3}{2}}\right] = \boxed{\frac{1}{27}\left[80\sqrt{10} - 13\sqrt{13}\right]\ \checkmark}$$

Arc Length Ex 2. Find the length of the parabola $y^2 = x$ between the points $(0,0)$ and $(1,1)$.




Use the formula for $x = g(y) = y^2$

$$\implies \frac{dx}{dy} = 2y \; and \; \sqrt{1 + \left(\frac{dx}{dy}\right)^2} = \sqrt{1 + 4y^2}$$

## Solve the integral by Trigonometric Substitution

And so, the arc length has the value

$$\boxed{L = \int_a^b \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy = \int_0^1 \sqrt{1 + 4y^2}\, dy}$$

- $y = \frac{1}{2}\tan(t) \Rightarrow dy = \frac{1}{2}\sec^2(t)dt$

  $\Rightarrow \sqrt{1 + 4y^2} = \sqrt{1 + \tan^2(t)} = \sec(t)$
- For the endpoints:

  $y = 0 \Longrightarrow t = 0 \; and \; 1 = y = \frac{1}{2}\tan(t) \Longrightarrow \alpha = arctan(2)$

Therefore, the arc length is

$$\boxed{L = \int_0^1 \sqrt{1 + 4y^2}\, dy = \int_0^{\alpha} \sec(t)\frac{1}{2}\sec^2(t)\, dt = \frac{1}{2}\int_0^{\alpha} \sec^3(t)\, dt}$$

## Arc Length: the last computation was made before

$$\boxed{L = \int_0^1 \sqrt{1+4y^2}\, dy = \tfrac{1}{2}\int_0^{\alpha} \sec^3(t)\, dt}$$

$$\boxed{= \frac{1}{4}\Big[\sec(t)\tan(t) + \ln\Big|\sec(t)+\tan(t)\Big|\Big]\Big|_0^{\alpha}}$$

$$\boxed{= \tfrac{1}{4}\Big[\sec(\alpha)\tan(\alpha) + \ln\Big|\sec(\alpha)+\tan(\alpha)\Big]}$$

Finally, since

$$\boxed{\tan(\alpha) = 2 \Longrightarrow \sec^2(\alpha) = 1+\tan^2(\alpha) = 5 \Longrightarrow \sec(\alpha) = \sqrt{5}}$$

Hence,

$$\boxed{L = \frac{\sqrt{5}}{2} + \frac{\ln(\sqrt{5}+2)}{4} \checkmark}$$

1. Given $f$ such that $f'$ is continuous on $[a, b]$, the Arc Length Function measures the arc length of the curve $y = f(x)$ from a particular starting point $(a, f(a))$ to any other point $(x, f(x))$ on the curve, for $a < x < b$:

$$\boxed{s(x) = \int_a^x \sqrt{1 + [f'(t)]^2}\, dt}$$

2. By the Fundamental Theorem of Calculus:

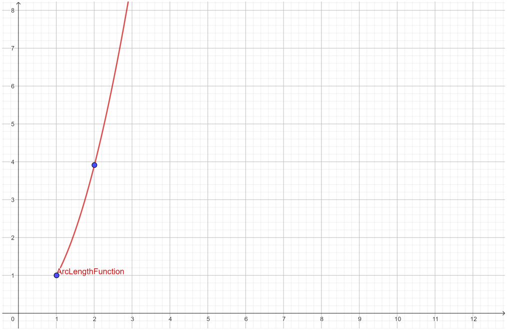
$$\boxed{\frac{ds}{dx} = \sqrt{1 + [f'(x)]^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \; and \; s = \int ds}$$

3. If $x = g(y) \Longrightarrow s(y) = \int_c^y \sqrt{1 + [g'(t)]^2}\, dt$ and

$$\boxed{\frac{ds}{dy} = \sqrt{1 + [g'(y)]^2} = \sqrt{1 + \left(\frac{dx}{dy}\right)^2}}$$

# Arc Length

Ex 3. a) Find the arc length function for the curve $y = x^2 - \frac{1}{8}\ln(x)$ starting at $P = (1, 1)$



Arc length function of the curve $y = x^2 - \frac{1}{8}\ln(x)$ starting at $P = (1, 1)$

Solution. • $f(x) = x^2 - \frac{1}{8}\ln(x) \Longrightarrow f'(x) = 2x - \frac{1}{8}\frac{1}{x}$, then

$$\bullet 1 + [f'(x)]^2 = 1 + \left(2x - \frac{1}{8x}\right)^2 \Longrightarrow \sqrt{1 + [f'(x)]^2} = \sqrt{1 + \left(2x - \frac{1}{8x}\right)^2}$$

$$= \sqrt{1 + 4x^2 - \frac{1}{2} + \frac{1}{64x^2}} = \sqrt{\left(2x + \frac{1}{8x}\right)^2} = \left(2x + \frac{1}{8x}\right)$$

• Hence, the arc length function is

$$\boxed{s(x) = \int_1^x \sqrt{1 + [f'(t)]^2}dt = \int_1^x \left(2t + \frac{1}{8t}\right)dt = t^2 + \frac{1}{8}\ln(t)\Big|_1^x}$$

$$\boxed{= x^2 + \frac{1}{8}\ln(x) - 1}$$

Ex 3. b) Find the arc length function for the curve $y = x^2 - \frac{1}{8}\ln(x)$ from $(1,1)$ to $(2,4)$

Solution. • The arc length function for any $x > 0$ is

$$\boxed{s(x) = \int_1^x \sqrt{1 + [f'(t)]^2} dt = \int_1^x \left(2t + \frac{1}{8t}\right) dt = t^2 + \frac{1}{8}\ln(t)\Big|_1^x}$$

$$\boxed{= x^2 + \frac{1}{8}\ln(x) - 1}$$

• The arc length function from $(1,1)$ to $(2,4)$ is then

$$\boxed{s(2) = 2^2 + \frac{1}{8}\ln(2) - 1 = 3 + \frac{\ln(2)}{8}}$$

## Arc Length for Parametrized Curves and Curves in Polar Coordinates

For a parametrized curve $(x(t), y(t))$ on $[a, b]$, the arc length is

$$\boxed{L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt}$$

For a curve in polar coordinates $(r(t)\cos(t), r(t)\sin(t))$ on $[\alpha, \beta]$ the arc length is

$$\boxed{L = \int_\alpha^\beta \sqrt{r^2(t) + \left(\frac{dr}{dt}\right)^2}\, dt}$$

Proof. Start with the known formula for a curve $y = f(x)$ and multiply and divide by $dt$: $\boxed{L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx}$

$$= \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, \frac{dx}{dt} dt = \int_a^b \sqrt{\left[1 + \left(\frac{dy}{dx}\right)^2\right] \left(\frac{dx}{dt}\right)^2}\, dt$$

$$= \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt \checkmark$$

Proof. Derivate

$$x(t) = r(t)\cos(t) \Longrightarrow \frac{dx}{dt} = x'(t) = r'(t)\cos(t) - r(t)\sin(t)$$

$$y(t) = r(t)\sin(t) \Longrightarrow \frac{dy}{dt} = y'(t) = r'(t)\sin(t) + r(t)\cos(t)$$

Then, inside the square root we obtain: $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2$

$$= [r'(t)\cos(t) - r(t)\sin(t)]^2 + [r'(t)\sin(t) + r(t)\cos(t)]^2$$

$$= r^2(t)\left(\cos^2(t)+\sin^2(t)\right) + \left(\frac{dr}{dt}\right)^2\left(\cos^2(t)+\sin^2(t)\right) \boxed{= r^2(t) + \left(\frac{dr}{dt}\right)^2}$$

Hence, the length is the integral of its square root:

$$\boxed{L = \int_\alpha^\beta \sqrt{r^2(t) + \left(\frac{dr}{dt}\right)^2}\, dt \ \checkmark}$$

Ex. Find the length of the curve given in polar coordinates by $r(t) = t^2$ on $[0, \sqrt{5}]$.

Solution. $\boxed{L = \int_0^{\sqrt{5}} \sqrt{r^2(t) + \left(\frac{dr}{dt}\right)^2}\, dt.}$

Here, $\boxed{r(t) = t^2 \Longrightarrow \frac{dr}{dt} = 2t}$, then the length is

$$L = \int_0^{\sqrt{5}} \sqrt{(t^2)^2 + (2t)^2} dt = \int_0^{\sqrt{5}} \sqrt{t^4 + 4t^2} dt = \int_0^{\sqrt{5}} \sqrt{t^2 + 4}\, t dt$$

Substitution for the antiderivative

$u = t^2 + 4 \Longrightarrow du = 2t\, dt \Longrightarrow$

$$\boxed{\int \sqrt{t^2 + 4} t\, dt = \frac{1}{2} \int \sqrt{u}\, du = \frac{1}{2}\frac{2}{3}(t^2 + 4)^{3/2}}$$

Hence

$$\boxed{L = \int_0^{\sqrt{5}} \sqrt{t^2+4}\, t\, dt = \tfrac{1}{3}(t^2+4)^{3/2}\Big|_0^{\sqrt{5}} = \frac{(9^{3/2} - 4^{3/2})}{3} = \frac{19}{3} \checkmark}$$